

Project Report: AI Registration Assistant

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Project Duration: 60+ Hours of Research & Implementation

Organization / Team: Data Alcott Systems

1. Executive Summary

The AI Registration Assistant is a full-stack, intelligent web application designed to streamline the university course registration process. Built over 60+ hours of intensive coding and research, the portal bridges the gap between complex administrative databases and student accessibility. By leveraging a multi-layered Large Language Model (LLM) architecture, the system provides an interactive conversational agent that allows students to view schedules, check seat availability, and register for courses using natural language, all while adhering to strict anti-hallucination guardrails.

2. Project Objectives

- **Automate Academic Advising:** Deploy an AI assistant capable of guiding students through course selection without manual administrative intervention.
- **Ensure Data Accuracy:** Implement strict agentic tools so the AI can only recommend courses that exist in the live database, eliminating AI hallucinations.
- **Secure Student Access:** Create a robust authentication flow requiring email-based One-Time Password (OTP) verification before account creation.
- **Centralize Administrative Control:** Provide a master admin dashboard to manage the course catalog, view chat logs, and analyze student distribution metrics.

3. System Architecture & Technology Stack

The application is built on a modern, high-performance tech stack, separating backend logic, AI orchestration, and frontend delivery into a seamless single-page application (SPA).

Backend & Database

- **FastAPI (Python):** Serves as the high-speed, asynchronous backbone of the application, handling API routing, security middleware, and payload validation via Pydantic schemas.
- **MongoDB:** A NoSQL database utilized to store dynamic course catalogs, student profiles, global system analytics, and historical chat logs.

AI Engine & Orchestration

- **Groq API:** Powers lightning-fast AI inference.
- **LangChain:** Orchestrates the AI agent, specifically binding Python functions (tools) to the LLM so it can read and write directly to the MongoDB database.

- **Multi-Model Fallback System:** The AI pipeline uses openai/gpt-oss-120b as the primary reasoning engine, with a 3-layer deep automatic fallback mechanism to smaller, faster models (like Qwen 27B) to ensure 100% uptime.
- **TextBlob:** Performs real-time sentiment analysis on student messages to flag negative interactions in the admin logs.

Security & Integrations

- **Brevo API:** Integrated via native Python urllib to dispatch real-world emails for OTP verifications and temporary password resets.
- **Cryptography:** Uses hashlib (PBKDF2-HMAC-SHA256) and secure salting to safely store student passwords.

Frontend UI

- **HTML5 & Tailwind CSS:** Delivers a responsive, highly styled user interface without the need for external frontend frameworks like React.
- **Chart.js & html2pdf.js:** Renders real-time data visualizations in the admin dashboard and allows for one-click PDF exports of analytics and chat logs.
- **Marked.js & MathJax:** Parses markdown and complex LaTeX mathematical formulas dynamically in the chat interface.

4. Key Features & Functionality

4.1. The Student Portal

- **Secure Registration:** Students must pass an email verification check (OTP) and a strict phone number validation check (rejecting dummy numbers) to register.
- **AI Chat Interface:** Students interact with an AI that knows their major and ID. The AI can execute tools to dynamically enroll them in courses (e.g., "Register me for CS101") or drop classes, instantly updating the database.
- **Multi-language Support:** Integrated Google Translate widget allows international students to navigate the portal in their native language.

4.2. The Administrator Console

- **Course Master Control:** Admins can add, edit, or delete courses. Time bounds are strictly validated (e.g., ensuring End Time is after Start Time).
- **Live Analytics:** A visual dashboard displaying total registrations, system logins, and a bar chart of student distribution across different academic fields.
- **Audit Logging & Exports:** Admins can view complete chat histories (including AI sentiment analysis) and export student records to CSV or dashboard metrics to PDF for offline review.